

SHTIP, North Macedonia

WMO: 135910

Lat: 41.750N

Lon: 22.183E

Elev: 1073

StdP: 14.14

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.2	20.1	8.6	8.9	17.0	13.5	11.4	22.8	33.9	51.8	29.3	39.0	5.0	320	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.3	96.3	69.1	93.3	68.7	90.5	67.8	71.3	90.5	70.1	88.9	69.0	86.9	6.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	97.1	76.3	64.1	93.1	75.6	62.9	89.4	75.2	35.7	90.6	34.7	88.9	33.7	87.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.9	24.0	20.5	DB	9.1	101.4	7.2	3.8	3.9	104.2	-0.3	106.5	-4.3	108.6	-9.5	111.4	
			WB	9.3	73.5	6.5	1.4	4.7	74.6	0.9	75.4	-2.8	76.2	-7.5	77.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	34.9	40.7	47.8	55.9	63.9	71.3	76.4	77.1	67.9	57.5	47.8	37.7
	DBStd	15.84	8.74	7.59	6.77	6.32	5.69	5.80	5.41	4.97	6.05	7.11	7.40	8.46
	HDD50	1419	472	268	122	20	0	0	0	0	0	21	126	390
	HDD65	4173	935	681	534	278	89	12	1	0	39	242	516	846
	CDD50	3851	3	7	53	197	432	639	819	840	536	255	61	9
	CDD65	1130	0	0	0	5	56	202	355	375	126	11	0	0
	CDH74	12245	0	0	1	66	531	2119	4123	4127	1147	130	1	0
	CDH80	5459	0	0	0	6	101	864	2031	2050	393	14	0	0
Wind	WSAvg	6.7	6.2	7.7	8.3	8.0	6.2	5.9	6.5	5.8	6.2	5.8	6.8	6.6
Precipitation	PrecAvg	21.0	1.4	1.2	1.5	2.0	2.1	1.9	1.4	1.4	1.8	2.2	1.9	2.2
	PrecMax	35.9	4.4	5.1	4.1	6.1	4.5	4.5	4.3	3.3	7.2	6.1	3.9	5.2
	PrecMin	12.5	0.0	0.1	0.1	0.2	1.0	0.5	0.0	0.1	0.2	0.0	0.4	0.0
	PrecStd	5.0	0.9	1.0	0.9	1.3	0.9	0.9	1.0	0.9	1.5	1.5	0.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	64.9	72.4	80.4	87.0	96.4	100.4	100.3	92.3	81.7	70.2	63.0
		MCWB	50.9	52.2	54.2	61.3	64.8	69.9	69.9	69.8	67.7	63.5	58.3	56.3
	2%	DB	54.4	60.1	67.9	75.3	82.8	92.3	96.2	96.7	88.5	77.7	65.7	57.0
		MCWB	48.5	49.7	52.6	58.8	64.1	69.4	69.1	69.1	66.0	61.1	56.9	51.7
	5%	DB	50.7	56.6	64.0	71.6	79.8	88.9	93.3	93.4	84.6	73.8	62.6	53.5
		MCWB	45.8	48.4	50.8	56.7	63.0	68.1	68.5	68.5	64.7	59.7	56.0	49.1
	10%	DB	47.5	53.2	60.2	68.1	76.7	85.4	90.3	90.3	80.8	69.9	59.8	50.2
		MCWB	43.7	46.5	49.0	54.9	61.9	66.8	67.9	67.5	63.4	58.8	53.7	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	53.8	56.3	62.8	67.5	72.8	72.8	72.9	70.3	65.1	60.1	56.4
		MCDB	57.5	62.8	69.0	77.0	83.1	91.6	93.6	93.6	87.2	77.9	66.9	62.1
	2%	WB	48.8	50.9	53.6	60.1	65.5	70.7	71.0	71.1	67.5	62.8	57.9	52.2
		MCDB	53.4	58.4	64.4	72.7	79.3	88.6	90.7	90.6	83.3	72.7	63.9	55.9
	5%	WB	46.3	49.0	51.9	57.9	64.0	69.1	69.9	69.9	65.9	61.2	56.2	49.9
		MCDB	50.1	55.7	61.7	69.2	77.2	85.8	89.0	89.0	80.0	70.3	61.9	53.0
	10%	WB	43.9	46.9	50.1	55.9	62.5	67.6	68.8	68.7	64.6	59.7	54.4	47.3
		MCDB	47.2	52.9	59.0	65.9	74.7	83.0	86.9	86.6	76.9	68.0	59.5	50.3
Mean Daily Temperature Range	5% DB	MDBR	11.9	16.0	18.3	19.0	20.5	22.3	24.2	24.3	21.4	19.1	14.5	11.6
		MCDBR	15.3	22.4	26.3	25.0	25.9	27.6	28.7	28.5	27.6	25.7	18.3	13.7
		MCWBR	10.5	13.4	13.5	11.7	10.4	9.4	9.1	9.0	10.4	12.0	10.6	9.4
	5% WB	MCDBR	13.1	20.0	22.9	22.7	23.4	26.0	25.8	25.9	23.2	20.7	15.9	11.6
		MCWBR	9.4	12.4	12.1	11.0	10.1	9.4	8.7	8.6	9.1	10.5	9.6	8.3
Clear-Sky Solar Irradiance	taub	0.332	0.359	0.404	0.462	0.454	0.438	0.436	0.452	0.431	0.394	0.362	0.335	
	taud	2.367	2.284	2.186	2.034	2.129	2.221	2.225	2.178	2.211	2.298	2.345	2.360	
	Ebn at Noon	255	265	265	257	262	267	266	257	253	249	240	242	
	Edh at Noon	27	34	42	52	49	44	44	44	40	33	27	25	
All-Sky Solar Radiation	RadAvg	529	807	1147	1474	1790	2057	2136	1902	1396	974	625	462	
	RadStd	81	104	133	168	117	153	124	124	112	112	62	77	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	-272	N/A	N/A

Nomenclature: See separate page